

WASP-52b

R-0525

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_180710 · created 2026-07-21T03:29:47+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2458309.8599 +/- 0.0042 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.21 +/- 0.062
Transit depth (Rp/Rs)^2	4.43 +/- 2.62 [%]
Orbital Inclination (inc)	89.79 +/- 2.83
Airmass coefficient 1 (a1)	1.5 +/- 0.8
Airmass coefficient 2 (a2)	-0.28 +/- 0.29
Scatter in the residuals of the lightcurve fit is	49.66 %
Best Comparison Star	None
Optimal Aperture	2.88
Optimal Annulus	5.75
Transit Duration (day)	0.0869 +/- 0.0085

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.21 ± 0.062 vs 0.1646 (deviation 0.73σ)
Duration fit vs published	0.0869 d vs 0.0758 d (deviation 1.3σ)
Mid-time O–C	-10.2 min
Depth SNR	3.4
Residual scatter	49.66 %

Light curve

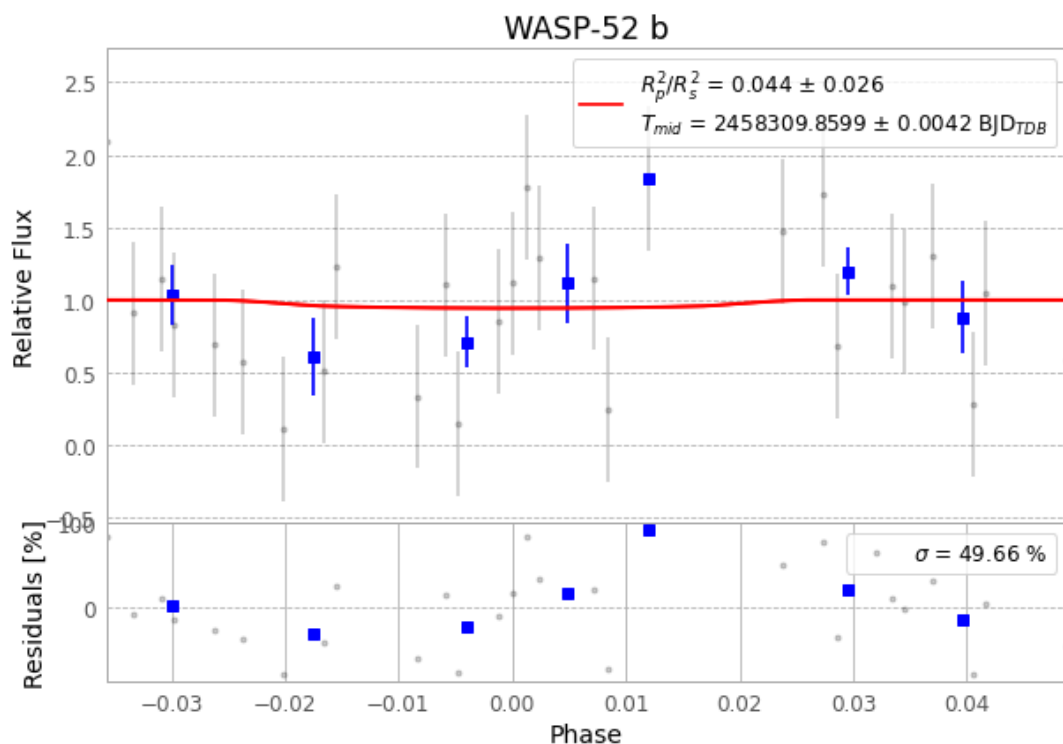


Figure 1 — FinalLightCurve_WASP-52 b_2018-07-10.png (SHA-256 efa010a1d2756587...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [561, 221], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 640a965397925a6b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 72f1d3a

Data provenance

73 FITS frames (48 MB), WASP-52180710070312.FITS → WASP-52180710103912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-180710.log (d8587eddd3f3...), FinalLightCurve_WASP-52 b_2018-07-10.png (efa010a1d275...), FinalLightCurve_WASP-52 b_2018-07-10.csv (880f34fbcfd2...)

Compute resources

Wall time 125.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-21 03:29Z.